Teacher Wu Kuan | See It at a Glance #7 — Why sin θ Never Exceeds 1

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1. The trap question

Why is the maximum of sin θ equal to 1?

2. The method

• On the unit circle, sin θ is the height of the point.
• Height ≤ radius = 1 → $-1 \le \sin\theta \le 1$.
• Max 1 at θ = 90°; min −1 at θ = 270°.

3. Parent check method

Ask your child: “Can a point on a circle of radius 1 be 1.2 units high?”

Practice

1. Can $\sin\theta = 1.2$?
2. Maximum value of $3\sin\theta$
3. Range of $\sin\theta + 2$
4. For $0° \le \theta < 360°$, when is $\sin\theta = 1$?
5. Minimum value of $5 - 2\sin\theta$

Answers

1. No
2. 3
3. 1 to 3
4. θ = 90°
5. 3

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